Prashant Mishra

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Professional Summary

Data Science Manager with 7+ years of experience in engineering production-grade systems including multi-agent conversational systems for enterprise financial institutions. Built ReAct-based agent architectures with dynamic tool routing, distributed LLM evaluation frameworks, and scalable RAG pipelines. Experienced in deploying GenAI workloads and partnering with client engineering teams to transition prototypes into secure, high-impact production systems.

Skills

- **GenAI:** RAG, Agentic AI, LLM Benchmarking and Selection, SLM fine-tuning, LLM-as-a-Judge, AI guardrails
- **Frameworks & Tools:** Python, LangChain, LangGraph, SQL, MCP servers, Arize AI, Terraform
- **Cloud & Platforms:** GCP, AWS, Azure, Databricks
- **Core Strengths:** AI Use case Discovery, Solution Architecture, LLMOps, Orchestrating Multi-agents frameworks

Professional Experience

Ernst & Young LLP | New York City, NY

Data Science Manager - Financial Services | 03/2023 - Present

Clients Served: Vanguard, Northern Trust, Edward Jones, Mastercard, First Citizens Bank (FCB), Visa

- Led engineering of a production multi-agent conversational platform (planner + retriever + tool-executor + synthesizer + evaluator agents) supporting 5 enterprise advisor workflows across financial institutions and reduced manual research time by ~60% and enabled over 1,000 daily advisor queries in pilot deployments.
- Implemented ReAct-style reasoning loops with dynamic tool routing, improving tool-call precision and decision making accuracy from ~60% to 85% for models with < 50B parameters through limited token usage, tool schema validation, optimized prompting using DSPy and iterative evaluation testing
- Built a distributed evaluation framework (LLM-as-Judge + rule-based validators) across 9 LLM scenarios with client focused datasets, benchmarking 12+ models and measuring latency, cost, and accuracy under load test which is used to drive model selection process for agents based on Tokens and Request limitations guiding over $10M in savings.
- Designed and implemented agent observability stack with structured logging and conversation-ID tracing across microservices using Arize AI, enabling root-cause debugging of tool failures, hallucinations, token usage and reducing production incident resolution time from ~4 hours to ~1 hour with automated alerting system
- Architected a multi-agent framework for real-time data retrieval (RAG) pipeline over 2M+ documents (PDFs, CRM notes, compliance memos) using semantic chunking of multi-model data (Chapter-subchapter relationship), and hybrid retrieval (vector search). Maintained accuracy at ~85% and optimized <3s P95 latency.
- Implemented AI safety and observability frameworks including grounding checks, bias detection, PII masking, and monitoring, reducing operational risk while meeting regulatory and compliance requirements in highly regulated financial environments.

- Automated deployment of GenAI services on AWS using containerized workloads in Prod, ENG and Test environments and CI/CD pipelines, provisioning IAM roles, API endpoints, and secure networking for regulated client environments.
- Served as a technical advisor in 15+ enterprise AI pursuits, partnering with executives to define AI adoption strategies, deliver executive workshops, identify high-value GenAI use cases, and shape multi-year AI transformation roadmaps.

Bill.com | San Jose, CA

Data Science Intern | 05/2022 - 08/2022

- Conducted exploratory data analysis using AWS Athena, Quicksight, Sagemaker, Tableau, and Python and validated the null hypothesis for customer retention. Preprocessed the data and experimented with the ML solution to improve the accuracy and recommendation of the model.

Amadeus Software Labs | Bangalore, India

Senior Software Engineer (Machine Learning) | 07/2018 - 07/2021

- Studied and explored 2 years of historical airport data to design and implement a passenger forecast tool to facilitate airport security and improve ground handling staff requirements.
- Analyzed the input dataset, conducted feature engineering, and included client-specific data points to achieve over 92% accuracy. Deployed the solution on the client server and set up model monitoring to estimate the data drift.

Research Experience

Research Assistant, NLP Lab | 08/2021 - 12/2022

- Conducted research studies in the field of NLP, focusing on topics of QA generation, Sentiment Analysis, and Named Entity Recognition, and published research papers in Medical Science leveraging NLP and Advanced Analytics.
- Papers: Audio delivery of health information: An NLP study of information difficulty and bias in listeners

Education

University of Arizona | Tucson, AZ

M.S. in Management Information Systems **|** 12/2022 | GPA : 3.9/4.0

Ramaiah Institute of Technology | India, Bangalore

B.S. Electronics and Telecommunications Engineering **|** 07/2018 | GPA : 3.9/4.0